

Linux Storage Management and LVM

Server Engineering Labs

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Topics Covered:

- Linux storage administration
- Partition management
- Logical Volume Management (LVM)
- Filesystem migration
- Persistent mounts with `fstab`

Environment:

Debian Linux Virtual Machine
Virtualized lab environment

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1. Introduction

This report documents several Linux storage administration tasks performed in a virtualized Debian environment. The main objective was to understand how Linux handles disks, partitions, mount points, and Logical Volume Management (LVM).

The exercises focused on:

- Detecting and managing storage devices
- Creating and mounting partitions
- Configuring LVM structures
- Migrating directories such as `/var`
- Making mounts persistent through `/etc/fstab`

The work was carried out in a controlled laboratory environment using virtual machines.

2. Disk Detection and Initial Inspection

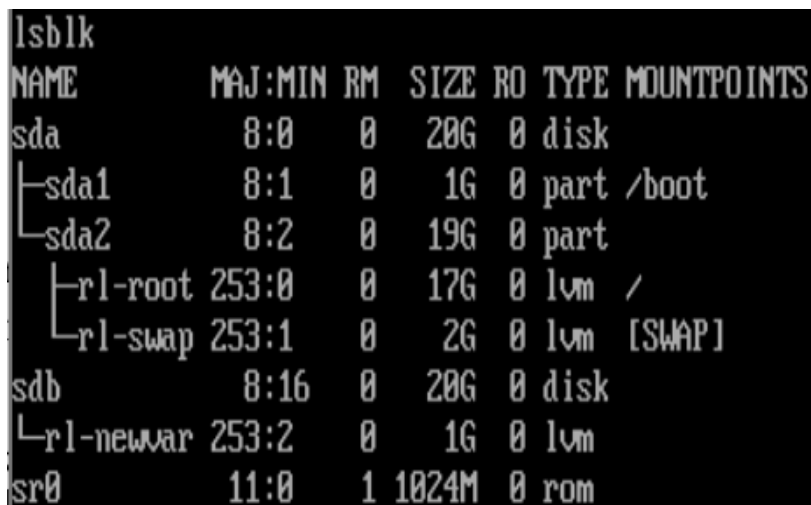
The first step consisted of inspecting the disks detected by the operating system.

Useful commands included:

```
lsblk
fdisk -l
df -h
```

These commands allow the administrator to:

- Identify physical disks
- Check existing partitions
- Verify mounted filesystems
- Inspect storage usage



```
lsblk
NAME            MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda              8:0    0   20G  0 disk
├─sda1           8:1    0    1G  0 part /boot
└─sda2           8:2    0   19G  0 part
   ├─rl-root     253:0    0   17G  0 lvm  /
   └─rl-swap     253:1    0    2G  0 lvm  [SWAP]
sdb              8:16    0   20G  0 disk
└─rl-newvar     253:2    0    1G  0 lvm
sr0             11:0    1 1024M  0 rom
```

Figure 1: Initial disk layout and LVM structure shown using `lsblk`.

3. Creating and Managing Partitions

A new partition was created in order to prepare additional storage for LVM.

Typical workflow:

```
sudo fdisk /dev/sdb
```

Inside `fdisk`, the partition table was modified and a new Linux partition was created.

After saving the changes, the partition table was reloaded:

```
sudo partprobe
```

The partition was then formatted:

```
sudo mkfs.ext4 /dev/sdb1
```

4. Logical Volume Management (LVM)

The laboratory also introduced the use of LVM for flexible storage administration.

The process included:

1. Creating a Physical Volume (PV)
2. Creating a Volume Group (VG)
3. Creating a Logical Volume (LV)
4. Formatting and mounting the LV

4.1. Creating the Physical Volume

```
sudo pvcreate /dev/sdb1
```

4.2. Creating the Volume Group

```
sudo vgcreate vg_data /dev/sdb1
```

4.3. Creating the Logical Volume

```
sudo lvcreate -L 5G -n lv_var vg_data
```

4.4. Formatting the Logical Volume

```
sudo mkfs.ext4 /dev/vg_data/lv_var
```

5. Migrating the /var Directory

One of the most important tasks was migrating the `/var` directory into the new logical volume.

The migration process required:

- Creating a temporary mount point
- Copying all files while preserving permissions
- Updating `/etc/fstab`
- Rebooting and validating the configuration

5.1. Temporary Mount

```
sudo mkdir /mnt/var_temp  
sudo mount /dev/vg_data/lv_var /mnt/var_temp
```

5.2. Copying Files

```
sudo rsync -av /var/ /mnt/var_temp/
```

```
CentOSRocky25-26 [Corriendo] - Oracle VM VirtualBox
cache db empty games lib lock lost+found nis preserve spool yp
[root@localhost var]# cd /
[root@localhost /]# rm -f /var/dormitorio
[root@localhost /]# ls /var
adm crash empty games lib lock lost+found nis preserve spool yp
cache db ftp kerberos local log mail opt run tmp
[root@localhost /]# umount /var
[root@localhost /]# ls /var
[root@localhost /]# touch /var/dormitorio
[root@localhost /]# ls -l /var
total 0
-rw-r--r--. 1 root root 0 Sep 22 21:23 dormitorio
[root@localhost /]# mount /var
[root@localhost /]# ls -l /var
total 84
drwxr-xr-x. 2 root root 4096 May 16 2022 adm
drwxr-xr-x. 8 root root 4096 Sep 22 17:59 cache
drwxr-xr-x. 2 root root 4096 May 16 2022 crash
drwxr-xr-x. 3 root root 4096 Sep 22 17:59 db
drwxr-xr-x. 2 root root 4096 May 16 2022 empty
drwxr-xr-x. 2 root root 4096 May 16 2022 ftp
drwxr-xr-x. 2 root root 4096 May 16 2022 games
drwxr-xr-x. 3 root root 4096 Sep 22 17:59 kerberos
drwxr-xr-x. 21 root root 4096 Sep 22 18:01 lib
drwxr-xr-x. 2 root root 4096 May 16 2022 local
lrwxrwxrwx. 1 root root 11 Sep 22 17:59 lock -> ../run/lock
drwxr-xr-x. 7 root root 4096 Sep 22 18:35 log
drwx-----. 2 root root 16384 Sep 22 19:23 lost+found
lrwxrwxrwx. 1 root root 10 May 16 2022 mail -> spool/mail
drwxr-xr-x. 2 root root 4096 May 16 2022 nis
drwxr-xr-x. 2 root root 4096 May 16 2022 opt
drwxr-xr-x. 2 root root 4096 May 16 2022 preserve
lrwxrwxrwx. 1 root root 6 Sep 22 17:59 run -> ../run
drwxr-xr-x. 6 root root 4096 Sep 22 17:59 spool
drwxrwxrwt. 4 root root 4096 Sep 22 20:59 tmp
drwxr-xr-x. 2 root root 4096 May 16 2022 yp
[root@localhost /]#
```

Figure 2: Validation of the temporary /var mount and filesystem behavior after unmounting and remounting operations.

5.3. Persistent Mount Configuration

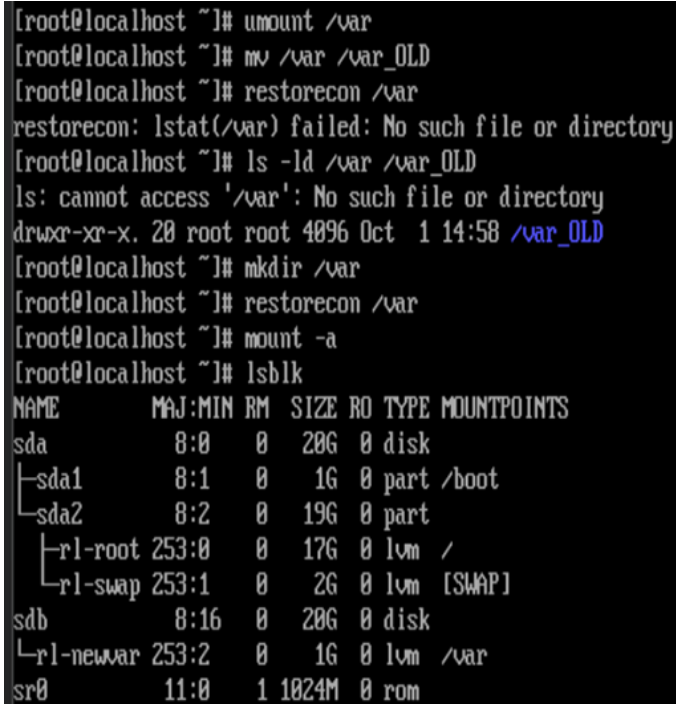
The filesystem was added to /etc/fstab:

```
UUID=<uuid> /var ext4 defaults 0 2
```

6. Verification and Validation

After rebooting the system, the configuration was verified using:

```
mount | grep var
df -h
lsblk
```



```
[root@localhost ~]# umount /var
[root@localhost ~]# mv /var /var_OLD
[root@localhost ~]# restorecon /var
restorecon: lstat(/var) failed: No such file or directory
[root@localhost ~]# ls -ld /var /var_OLD
ls: cannot access '/var': No such file or directory
drwxr-xr-x. 20 root root 4096 Oct  1 14:58 /var_OLD
[root@localhost ~]# mkdir /var
[root@localhost ~]# restorecon /var
[root@localhost ~]# mount -a
[root@localhost ~]# lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	20G	0	disk	
└sda1	8:1	0	1G	0	part	/boot
└sda2	8:2	0	19G	0	part	
└└rl-root	253:0	0	17G	0	lvm	/
└└rl-swap	253:1	0	2G	0	lvm	[SWAP]
sdb	8:16	0	20G	0	disk	
└rl-newvar	253:2	0	1G	0	lvm	/var
sr0	11:0	1	1024M	0	rom	

Figure 3: Final LVM configuration after mounting the logical volume on `/var`.

The logical volume was successfully mounted and persisted correctly after reboot.

7. Challenges Encountered

Several issues appeared during the lab:

- Incorrect mount configurations
- Filesystem permission inconsistencies
- Risks when modifying `/etc/fstab`

Particular care had to be taken because incorrect `fstab` entries can prevent Linux from booting correctly.

8. Conclusion

This laboratory provided practical experience with Linux storage administration and LVM management.

The exercises demonstrated:

- How Linux organizes storage devices
- How logical volumes improve flexibility

- How to safely migrate important directories
- The importance of persistent filesystem configuration

These concepts are essential for Linux system administration, server deployment, and infrastructure management.